

Annotation rules – Siret3

Version 1.0 · 25 September 2026 · How to annotate the challenge tiles in Marcaj. The reference used for scoring follows exactly these rules.

■ AT A GLANCE

LABEL	GEOMETRY	WHAT IT IS	ATTRIBUTES
vineyard	polygon	the canopy of one grapevine plant, seen from above	vineyard_id
waste	bounding box	one piece of litter, or one inseparable cluster of litter	vineyard_id
row	polyline	the centre line of one vine row	vineyard_id, row_id, row_structure
interrow_area	polygon	the ground between the canopies of two neighbouring rows	vineyard_id, interrow_cover

row_structure = regular / disrupted / unassessable · interrow_cover = bare_soil / vegetation / mixed / unassessable. Label and attribute names must be written exactly as here, in lower case.

GOLDEN RULES

1. **One canopy = one plant.** Never one polygon over a whole row.
2. **Grapevines only.** Fruit trees, shrubs, weeds and grass are not vineyard.
3. **A row is one line per tile,** from its first to its last vine, through any gaps.
4. **Canopies and inter-row areas never overlap.** The inter-row polygon runs from canopy edge to canopy edge, not from axis to axis.
5. **IDs survive tile edges:** the same block and the same row keep their vineyard_id and row_id in every tile they appear in.
6. **Every tile gets an answer:** annotate it, or mark it as having no objects — and submit the job.

■ LABELS AND ATTRIBUTES, IN PLAIN WORDS

Every object you draw has a **label** and a few **attributes**. The label says *what the object is* — a canopy, a row, a piece of waste. Attributes are *the extra facts about that particular object*, like the fields of a form you fill in for it: which block it belongs to, which row it is, what state it is in.

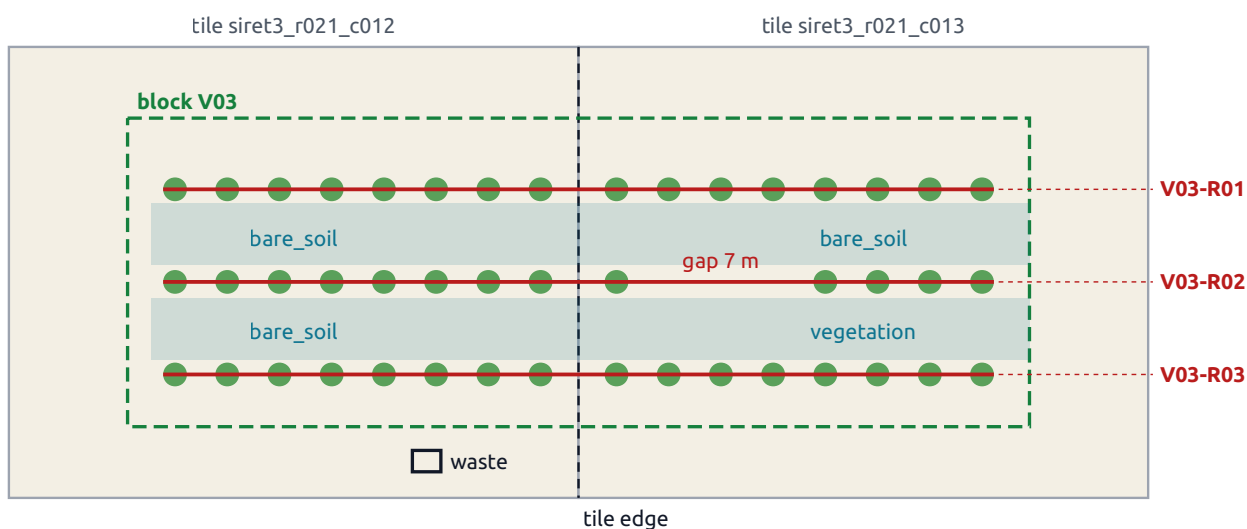
You set them in Marcaj when an object is selected: the attribute panel next to the canvas shows the fields of its label. Fields with a fixed list of answers are a drop-down; ID fields are free text. Pre-annotations imported from your model carry their attributes along, so your model can fill them in and you only correct them.

ATTRIBUTE	ON	WHAT IT TELLS	WHAT IT IS USED FOR
vineyard_id	every object	which vineyard block the object belongs to — the block's name, e.g. V03	Groups objects into blocks. The number of different vineyard_id values is the block count ; whether the right objects share one ID is the grouping score.
row_id	row	which physical row this polyline is — its name, e.g. V03-R02	Ties together the pieces of one row that lie in different tiles. The number of different row_id values is the row count .

ATTRIBUTE	ON	WHAT IT TELLS	WHAT IT IS USED FOR
row_structure	row	is the row continuous: <code>regular</code> , <code>disrupted</code> or <code>unassessable</code>	A classification scored against the reference (section 4.3). Tells the grower where vines are missing.
interrow_cover	interrow_area	what covers the ground: <code>bare_soil</code> , <code>vegetation</code> , <code>mixed</code> or <code>unassessable</code>	A classification scored against the reference (section 5.2). Tells the grower which inter-rows need mowing or tilling.

The ID values are your own. You may call a block `V03`, `north` or `7` — the reference may call it something else, and that is fine. What is scored is that *the same block always has the same ID*, different blocks have different IDs, and each row has one ID of its own.

Worked example: one block, three rows, two tiles



Block `V03` crosses the edge between two tiles. Its three rows are cut by the edge, so each row is drawn **twice** — **once per tile** — **with the same** `row_id`. Row `R02` has a 7 m gap, but only in the right tile. The objects and their attributes:

TILE	LABEL	VINEYARD_ID	ROW_ID	ROW_STRUCTURE	INTERROW_COVER
r021_c012	row	V03	V03-R01	regular	—
r021_c012	row	V03	V03-R02	regular	—
r021_c012	row	V03	V03-R03	regular	—
r021_c012	interrow_area ×2	V03	—	—	bare_soil, bare_soil
r021_c012	vineyard ×24	V03	—	—	—
r021_c012	waste	V03	—	—	—
r021_c013	row	V03	V03-R01	regular	—
r021_c013	row	V03	V03-R02	disrupted	—
r021_c013	row	V03	V03-R03	regular	—
r021_c013	interrow_area ×2	V03	—	—	bare_soil, vegetation
r021_c013	vineyard ×21	V03	—	—	—

Result: **1 block** (one `vineyard_id`), **3 rows** (three `row_id` values, although six polylines were drawn), 45 canopies. Had the right tile's rows been named `R04`–`R06`, the count would read 6 rows — a wrong answer.

The same row in `annotations.xml` of the pre-annotation ZIP:

```
<polyline label="row" points="112.0,1830.5;1990.4,402.7" occluded="0">
  <attribute name="vineyard_id">V03</attribute>
  <attribute name="row_id">V03-R02</attribute>
  <attribute name="row_structure">disrupted</attribute>
</polyline>
```

Common attribute mistakes

MISTAKE	WHAT IT COSTS
Rows renumbered from 1 in every tile	One physical row counts as several: the row count is wrong, and rows in different tiles look like different rows.
A new <code>vineyard_id</code> for the same block in each tile	The block count is wrong and the grouping score drops.
An attribute left empty	A missing <code>row_structure</code> or <code>interrow_cover</code> is a wrong answer; an object without <code>vineyard_id</code> belongs to no block.
Values typed differently: Regular, bare soil, grass	Not one of the allowed values — the answer is wrong. Use the drop-down, or the exact lower-case names in your model's output.
Attributes set on the wrong label, e.g. <code>interrow_cover</code> on a row	Ignored. Cover belongs to <code>interrow_area</code> , structure to <code>row</code> .

■ 1. BEFORE YOU START

- **Tiles.** 2048 × 2048 px GeoTIFF, 2.5 cm per pixel (51.2 m on the ground), EPSG:32635. Annotate in pixels in Marcaj — the georeferencing is kept by the platform, and scoring converts to metres.
- **Annotate every tile you were given.** Scoring uses a hidden subset of them, including tiles with no vineyard at all.
- **Tiles with nothing to annotate:** tick *No objects in this frame*. A tile with no answer blocks the job from being submitted.
- **Only submitted jobs are scored.** A saved but unsubmitted job counts as not annotated.
- **Pre-annotations** come in with the tiles as one CVAT for images 1.1 ZIP (Appendix A). The labels and attributes of your model's output must use the names above.

1.1 Order of work and teamwork

The order in which you annotate the tiles does not matter for scoring: the reference is matched to each tile by its file name, and any tile may be in the scored subset. Three things do matter:

1. **Upload all five parts before you publish.** Pre-annotations can be imported only into an unpublished project. Run your model over all 311 tiles, upload all five ZIPs, check that the project has 311 files — and only then publish. After publishing, no more pre-annotations can be imported.
2. **Agree on IDs before splitting the work.** A block or a row that crosses a tile edge keeps one `vineyard_id` / `row_id` in every tile. If neighbouring tiles are annotated by different people who number on their own, the block and row counts come out wrong. The simplest way: let your model assign the IDs for the whole orthomosaic at once, and have people only correct them.
3. **Take neighbouring tiles together.** File names are `siret3_r<row>_c<column>`, and the ZIPs keep them in name order, so tiles next to each other in a grid row sit next to each other. One person annotating a patch of neighbouring tiles has the fewest mismatches at the tile edges.

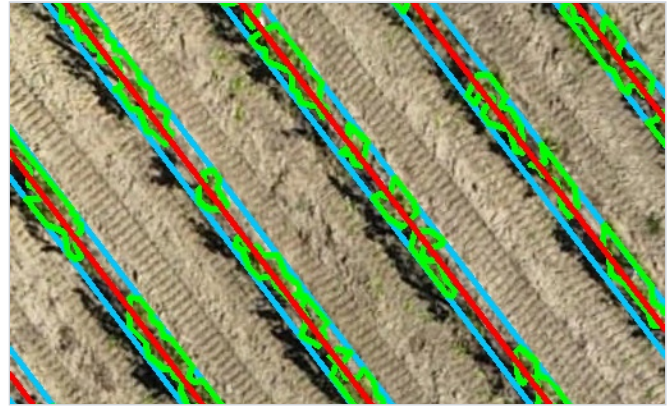
■ 2. GRAPEVINE CANOPIES — VINEYARD

A polygon around the foliage of one grapevine plant, as seen from above. Trace the outline of the leaves, within about 10 cm. Leave out bare soil, the plant's shadow, weeds and grass around it.

2.1 Young vines on bare soil



What you see: young vines, a few leaves each, in rows 2.7 m apart on tilled soil. The dark streaks next to them are shadows.



Annotated: each plant a small green polygon, the row axis in red through the plants, the inter-row areas (cyan) between the canopy edges.

- Each plant is its own polygon, however small. Leaf clumps smaller than about 0.2 m² that are not part of a plant (weeds) are not annotated.
- White protective tubes and stakes next to young vines belong to the planting — they are neither canopy nor waste.

2.2 Older vines with touching canopies

- Where neighbouring plants touch, **split the foliage between them** at the point where it visibly narrows.
- If the canopy is continuous with no visible narrowing, split it at the in-row planting distance — measured from visible trunks, stakes or the gaps elsewhere in the same row, typically 1.0–1.5 m.
- Never draw one polygon over a whole row or a whole block: the canopy count is scored.

2.3 Trees, orchards and other vegetation



NOT A VINEYARD An orchard: fruit trees are also planted in rows, but their crowns are 2–4 m wide, round, and 4–6 m apart. Vines are under 1 m wide and planted every 1–1.5 m along the row.



NOT A CANOPY A tree standing inside a vineyard. Do not annotate it as `vineyard`. The vines around it are annotated as usual; where it hides the row, see 4.3.

- Not `vineyard`: fruit and forest trees, shrubs, hedges, weeds, grass, crops, vines climbing on fences, arbours or houses.
- Garden vineyards in the village outskirts are annotated if the vines stand in at least three rows; a single vine or an arbour in a yard is not.

2.4 Shadows, tile edges, missing plants

- **Shadow**: trace the leaves, not their shadow. A plant lying entirely in a deep shadow where it cannot be seen is not drawn.
- **Tile edge**: a plant cut by the tile edge is traced up to the edge; its other part is a separate polygon in the neighbouring tile.

- **Missing or dead plant:** nothing is drawn — the gap shows in the row's `row_structure` (4.3).

■ 3. WASTE — `WASTE`

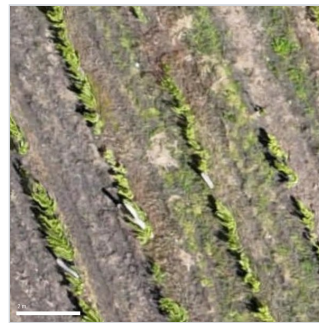
A tight axis-aligned box around a clearly visible piece of litter, within the vineyards and the land around them (anywhere on the tile). The box area is not taken as the waste's area.



WASTE A white bag at the edge of a vineyard. One box, tight around the object.



WASTE Two separate objects next to each other: two boxes. If they overlapped and could not be told apart, one box around both.



NOT WASTE White plastic tubes and stakes protecting young vines. They are part of the planting.



NOT WASTE A pale patch of bare or dry soil between rows. Waste is an object that does not belong to the ground.

WASTE	NOT WASTE
bags, plastic sheets and film, bottles, cans, packaging, tyres, construction debris, heaps of rubbish	vine tubes, stakes, trellis posts and wires, irrigation hoses, stones, bare or pale soil, flowering shrubs, pruning residue and cut branches, vehicles and machinery

- **Separate objects** get separate boxes. An **inseparable cluster** — pieces lying on top of each other — gets one box.
- `vineyard_id`: the block the object lies in, or the nearest block within 10 m. Farther away, leave it empty.
- When in doubt whether something is litter, leave it out: a false box costs as much as a missed one.

■ 4. ROW AXES — `ROW`

A polyline along the centre line of one row of vines. The axis is scored against the reference with a 0.4 m tolerance and 80% mutual coverage.

4.1 How to draw

- **One polyline per physical row per tile**, from the first to the last vine of the row in this tile, or to the tile edge.
- Keep the line within 0.2 m of the centre of the vines. For a straight row two points are enough; for a curved row add a point where it bends.
- **Gaps do not split a row.** The polyline runs straight through missing plants, and the row keeps its `row_id`.
- The outermost rows of a block are annotated like any other.

4.2 `row_id`

- Unique within its block and **the same in every tile the row crosses**. A row cut by a tile edge is two polylines with one `row_id`.
- Any text is accepted; we recommend `<vineyard_id>-R<nn>`, e.g. `V03-R017`, numbered across the block.
- The number of distinct `row_id` values is scored as the row count — a row renumbered in each tile counts twice.

4.3 `row_structure`

VALUE	WHEN
regular	The row is continuous in this tile: no gap of 5 m or more.

VALUE	WHEN
disrupted	A visible gap of 5 m or more along the row in this tile — several missing or dead vines in a row, or a tree or obstacle standing in the row.
unassessable	The row cannot be made out over most of its length in this tile: deep shadow, overexposure, weeds taller than the vines. This is the exclusion value, compared with the reference like any other answer.

The value describes the row **within this tile**: the same row may be `regular` in one tile and `disrupted` in the next.

4.4 Where rows end



Rows running along a dirt road. The road is not part of the block: rows and inter-row areas stop at the edge of the planting, not at the road centre.

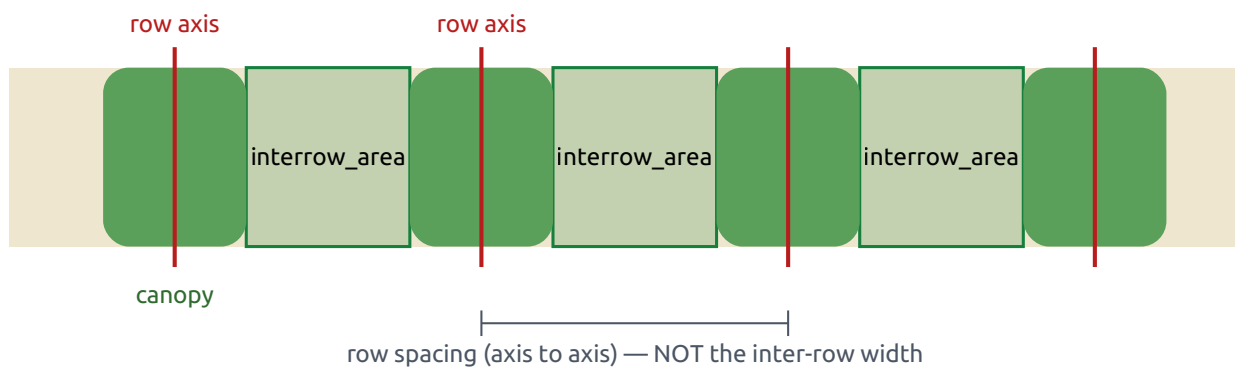


The edge of a block against scrub. The last row is annotated; there is no inter-row area outside it.

■ 5. INTER-ROW AREAS — `INTERROW_AREA`

A polygon of the ground between the canopies of two neighbouring rows of the same block. It is the walkable ground the route uses, and its area is scored.

5.1 How to draw



- **Long sides:** along the canopy edges of the two rows. The canopies themselves are not part of it.
- **Short sides:** where the rows end — the headland beyond the row ends, roads and exterior land are not included. If one row is shorter, end at the shorter one.
- **Holes:** cut out trees and buildings standing in the inter-row.
- **One polygon per inter-row per tile**, cut at the tile edge. None outside the outermost rows of a block.

5.2 interrow_cover



bare_soil — tilled soil, vegetation covers less than about a quarter of the inter-row.



mixed — strips of grass and weeds alternate with bare soil; between a quarter and three quarters covered.



vegetation — grass or weeds cover more than about three quarters of the inter-row.

unassessable: the ground cannot be seen — deep shadow, or a canopy closed over the inter-row. Judge each inter-row on its own, within this tile.

■ 6. BLOCKS — VINEYARD_ID

- A **vineyard block** is a connected planting area. Two plantings are the same block when they touch or are separated by less than 5 m of non-vineyard ground. **A road or a track always separates blocks.**
- Every object — canopy, row, inter-row area, and waste within 10 m of a block — carries the `vineyard_id` of its block.
- The same block keeps the same `vineyard_id` in every tile. Any text is accepted (we recommend `V01`, `V02` ...); only the grouping is compared with the reference, not the values.
- The number of distinct `vineyard_id` values is scored as the block count.

■ 7. WHAT YOU DO NOT ANNOTATE

- Inter-row axes — only the inter-row *areas*.
- Roads, tracks, passages, forbidden zones and inspection targets — the organizers supply passages, forbidden zones and the starting point as GeoJSON; your inspection targets go into your application's output, not into Marcaj.
- Orchards, trees, shrubs, buildings, fences, crops other than grapevine.
- Anything in the black area outside the imagery on edge tiles.

■ 8. CHECKLIST BEFORE SUBMITTING

- No polygon covers more than one plant; no canopy on trees or orchard crowns.
- Every row has one polyline per tile, straight through its gaps, with the same `row_id` on both sides of each tile edge.
- No inter-row area overlaps a canopy or extends past the row ends.
- Every object has `vineyard_id`; every row has `row_id` and `row_structure`; every inter-row area has `interrow_cover`.
- Every tile is annotated or marked *No objects in this frame*, and every job is **submitted**.

■ APPENDIX A. UPLOAD ZIP

One ZIP in the **CVAT for images 1.1** format, uploaded to your Marcaj project **before it is published**. Keep each ZIP under 90 MB (upload in several parts if needed). The tiles must be the supplied files, unchanged and with their original names; JPEG/PNG are not accepted.

```
team_upload.zip
├── annotations.xml
├── images/
│   └── siret3_r021_c012.tif ...
```

Label definitions for annotations.xml (<meta><task><labels>):

```
<label><name>vineyard</name><type>polygon</type><attributes>
  <attribute><name>vineyard_id</name><input_type>text</input_type></attribute></attributes></
label>
<label><name>waste</name><type>rectangle</type><attributes>
  <attribute><name>vineyard_id</name><input_type>text</input_type></attribute></attributes></
label>
<label><name>row</name><type>polyline</type><attributes>
  <attribute><name>vineyard_id</name><input_type>text</input_type></attribute>
  <attribute><name>row_id</name><input_type>text</input_type></attribute>
  <attribute><name>row_structure</name><input_type>select</input_type>
    <values>regular
disrupted
unassessable</values></attribute></attributes></label>
<label><name>interrow_area</name><type>polygon</type><attributes>
  <attribute><name>vineyard_id</name><input_type>text</input_type></attribute>
  <attribute><name>interrow_cover</name><input_type>select</input_type>
    <values>bare_soil
vegetation
mixed
unassessable</values></attribute></attributes></label>
```

The example tiles in the Assets are annotated by these rules and packed in exactly this format — open them in Marcaj to see what a finished tile looks like.